The Feedback Systems

Department of Medical Physiology and Microbiology

College of Medicine

Hawler Medical University

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Learning Outcomes:

- Define Homeostasis.
- Describe the components of a feedback system.
- Contrast the operation of negative and positive feedback system.
- Explain homeostatic imbalances.

- Humans have many ways to maintain Homeostasis.
- Homeostasis is the state of relative stability of the body’s internal environment.
- Feedback systems are corrective cycles that help restore the conditions needed for health and life.

- Disruptions to homeostasis often set in motion feedback systems.

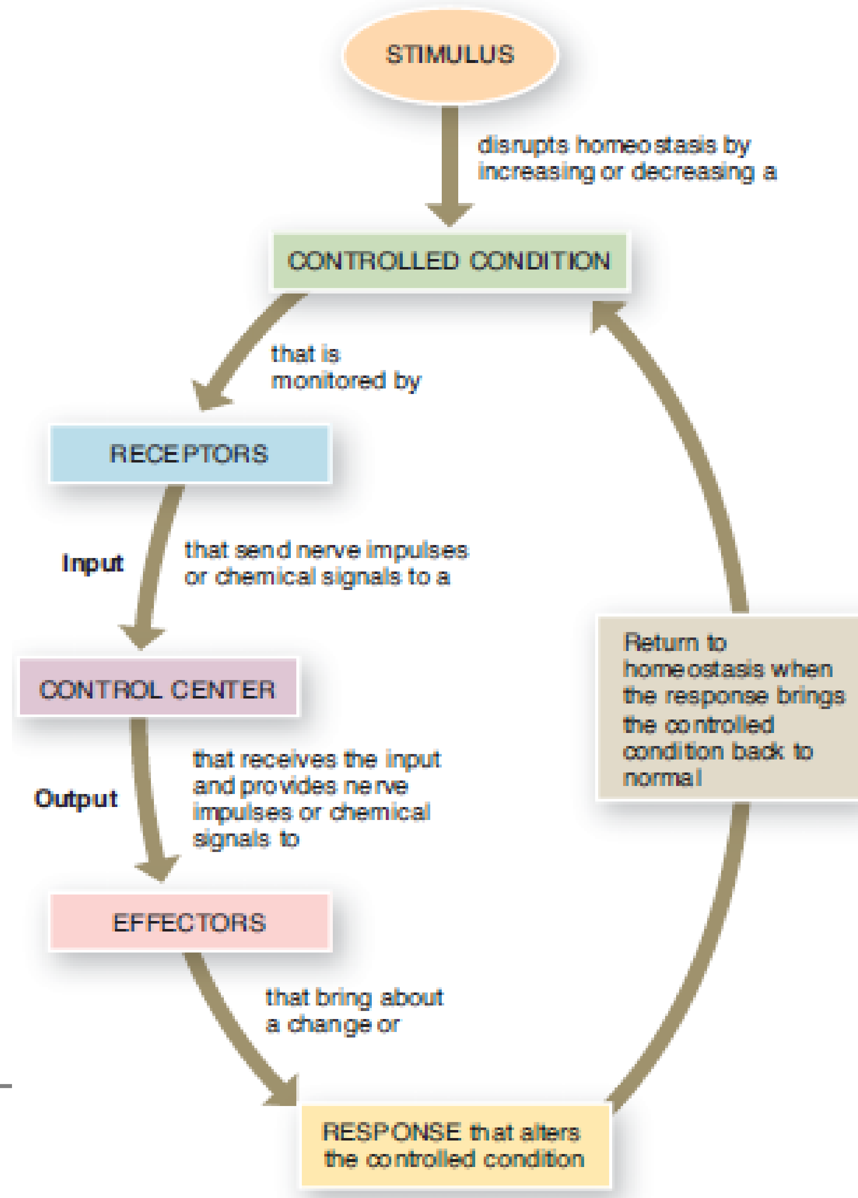
- Human body have many ways to maintain homeostasis, the state of relative stability of the body’s internal environment.

Feedback System

- Cycle of events
  - Body is monitored and re-monitored.
  - Each monitored variable is termed a controlled condition e. g. Blood pressure, temperature and blood glucose level et c.
  - Any disruption that changes a controlled condition is called a stimulus.

Three Basic components:

- Receptor
- Control center
- Effector



Feedback Systems

- **Receptor**
  - Body structure that monitors changes in a controlled condition
  - Sends **input** to the control center

- **Receptor**
  - This pathway is called an **afferent pathway** since the information flows toward the control center.
  - Typically, the input is in the form of nerve impulses or chemical signals.
    - Nerve ending of the skin in response to temperature change .

Feedback Systems

- **Control Center**
  - **Brain**
  - Sets the range of values to be maintained
  - Evaluates **input** received from receptors and generates **output** command
    - E.g: Brain acts as a control center receiving nerve impulses from skin temperature receptors

- **Control Center**
  - **Brain**
  - This pathway is called an efferent pathway since the in formation flows away from the control center.
  - Output from the control center typically occurs as nerv e impulses, or hormones or other chemical signals

Feedback Systems

- **Effector**
  - Receives output from the control center
  - Produces a response or effect that changes the c ontrolled condition
    - Found in nearly every organ or tissue
    - When body temperature drops sharply the brain sends a nd impulse to the skeletal muscles (effecter) to contract
    - Shivering to generate heat

Negative and Positive Feedback systems:

- A feedback system. can regulate a controlled condition in the body's internal environment.

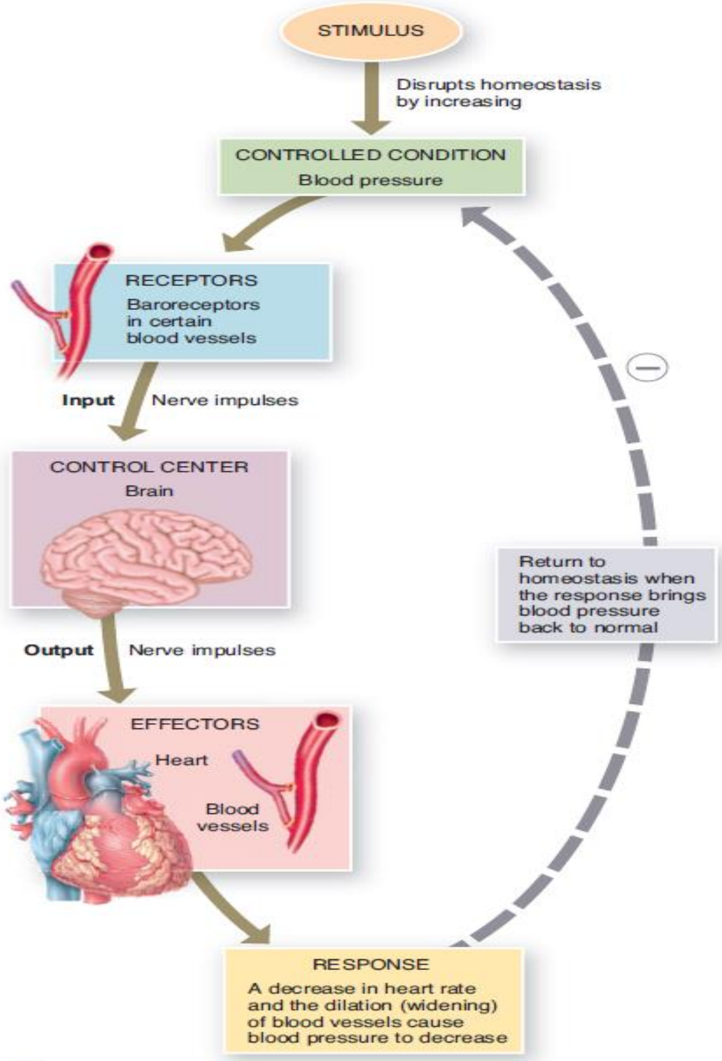
- In a feedback system, the response of the system "feeds back" information to change the controlled condition in some way, either negating it (negative feedback) or enhancing it (positive feedback)

Negative and Positive Feedback systems

- **Negative Feedback systems**
  - Reverses a change in a controlled condition
    - Regulation of blood pressure (force exerted by blood as it presses again the walls of the blood vessels)
- **Positive Feedback systems**
  - Strengthen or reinforce a change in one of the body's controlled conditions
    - Normal child birth

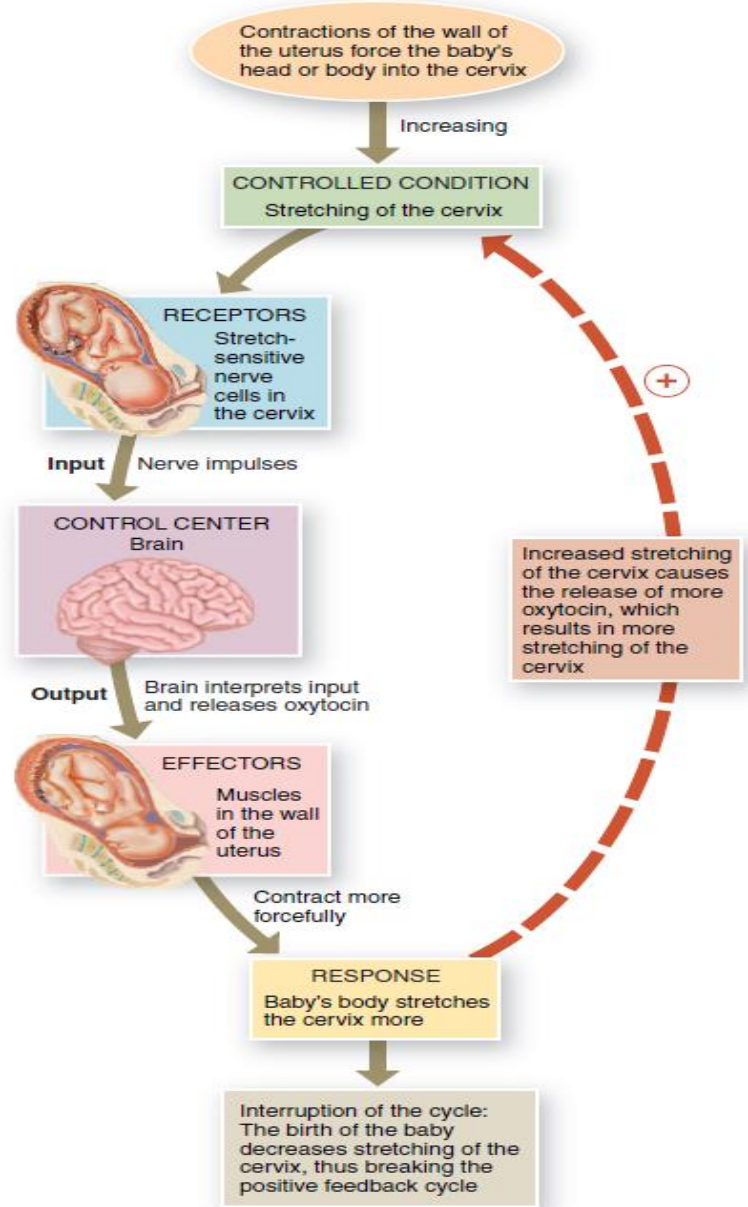
Negative Feedback: Regulation of Blood Pressure

- External or internal stimulus increase BP
  - Baroreceptors (pressure sensitive receptors)
    - Detect higher BP
    - Send nerve impulses to brain for interpretation
    - Response sent via nerve impulse sent to heart and blood vessels
    - BP drops and homeostasis is restored
    - Drop in BP negates the original stimulus



Positive Feedback Systems: Normal Child birth

- Uterine contractions cause va gina to open
- Stretch-sensitive receptors in cervix send impulse to brain
- Oxytocin is released into the blood
- Contractions enhanced and b aby pushes farther down the uterus
- Cycle continues to the birth of the baby (no stretching)



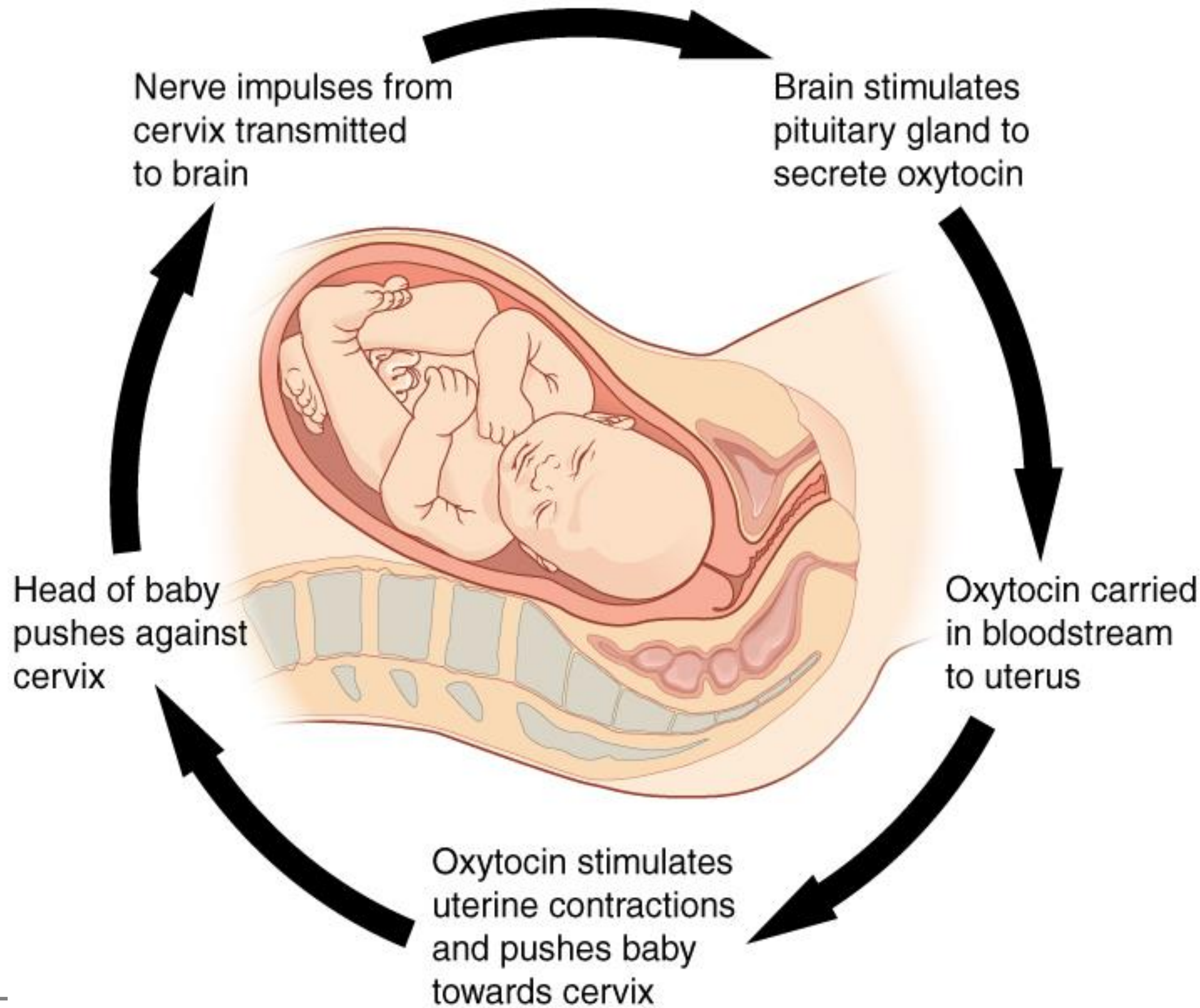
Homeostatic Imbalances:

- The many factors in maintaining Homeostasis or Health include the following:

- The environment and your own behavior.
- Your genetic makeup.
- The air you breathe, the food you eat, and even the thoughts you think.

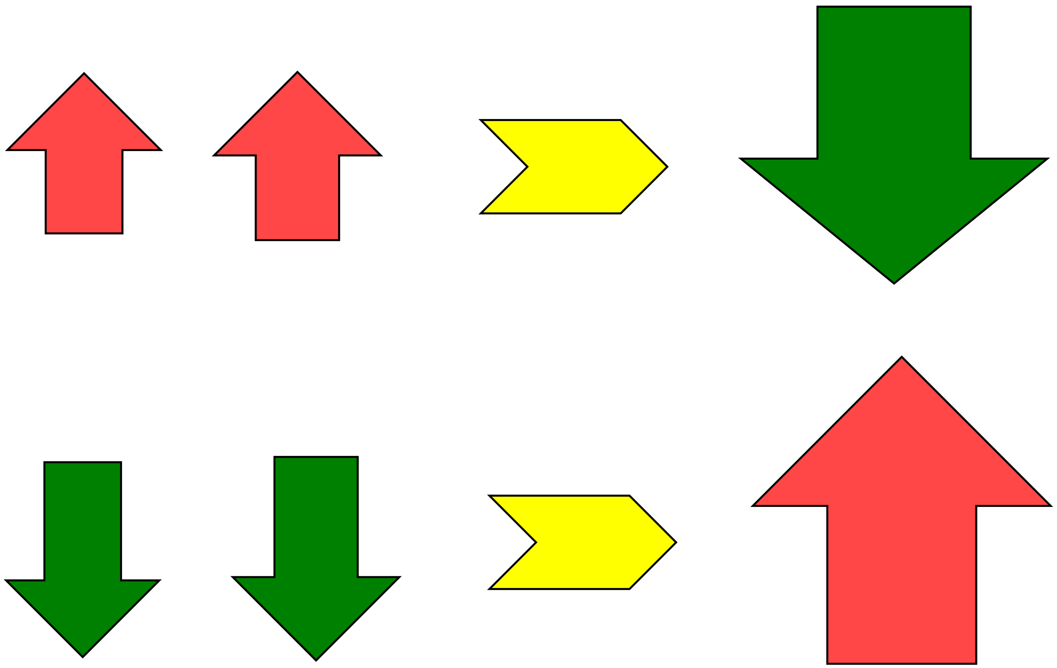
Homeostatic Imbalances

- Normal equilibrium of body processes are dis rupted
  - **Moderate imbalance**
    - Disorder is any abnormality of structure and function
    - Disease specific for an illness with recognizable signs a nd symptoms
    - **Signs** are objective changes such as a fever or swelling
    - **Symptoms** are subjective changes such as headache
  - **Severe imbalance**
    - Death


Nerve impulses from cervix transmitted to brain
Brain stimulates pituitary gland to secrete oxytocin
Oxytocin carried in bloodstream to uterus
Oxytocin stimulates uterine contractions and pushes baby towards cervix
Head of baby pushes against cervix

Negative Feedback System

- When a decrease in a stimulus ends with increase or an increase results in a decrease



Positive Feedback System

- When a decrease in a stimulus ends with more decrease or an increase results in more increase